Recovering the dynamics of a sequence seen only through noise

Results in brief — 25 September 2026

1. What we are doing, and why this way

The problem. A diffusion model generates data by learning to reverse a noising process. To reverse it, it must learn one object at every noise level: the *score*, the direction in which noisy data becomes more likely. On real data nobody knows the true score, so nobody can measure how far a learned one is from it — progress is judged by proxies such as sample quality. That leaves a class of questions unanswerable. How much information does the noise actually destroy? What kind of structure is lost first? And does knowing something about how the data is built help, or is a large enough network always enough?

The trick. We work where the true score can be computed exactly, so those questions have exact answers. The data comes from a Markov chain: a sequence $a_1, \ldots, a_L$ in which each site is a fraction of the previous one plus fresh randomness,

$$a_{i+1} \;=\; \rho\, a_i + \varepsilon_i .$$

Because the dependencies form a chain, the exact score of the *noised* sequence can be computed by message passing, to numerical precision. Every error reported below is measured against that truth rather than against a proxy.

The two knobs. ρ is the correlation: how strongly each site remembers the last. The innovation ε is drawn from a *generalised Gaussian*, whose density falls off like $e^{-|x/s|^{\beta}}$. The exponent β is the **shape**: it sets how the randomness is distributed at a fixed spread. The scale s is always chosen so the variance is identical for every β, so β moves the *fourth* moment while the second stays put — which is exactly what separates "non-Gaussian structure" from "more noise".

β	distribution	tails, against Gaussian	excess kurtosis
1	Laplace	heavier: occasional large jumps	+3.00
1.5	—	slightly heavier	+0.76
2	Gaussian	the reference	0.00
3	—	lighter: large jumps suppressed	−0.58

Table 1: The shape coordinate β, at fixed variance. Excess kurtosis is $\Gamma(5/\beta)\Gamma(1/\beta)/\Gamma(3/\beta)^2 - 3$, zero exactly at the Gaussian. The main chain uses $\beta = 1$; one experiment uses a chain on the *negative* side, to check that what matters is fourth-order structure rather than heavy tails specifically.

The comparison. Two ways of learning the score are put side by side. Ours knows the data is a chain, and fits the chain's parameters by EM with exact message passing. The competitors are neural networks — transformers and convolutional window models — that know nothing about the structure. Both are shown *exactly the same information*: one noisy view of each sequence, never the clean one, and neither may consult the true score to decide when to stop training. That last condition is the one comparisons of this kind most often violate.

How we measure it. Every number in this document is a *relative score error*: the size of the gap between a method's score and the true one, divided by the size of the true score itself, pooled over twelve noise levels. It has three natural landmarks. The exact score scores 0. Ignoring the data's structure entirely — answering as if the sites were independent — scores 0.670. And 0.168 is the best score achievable by any method that captures correlations but nothing beyond them; we prove that this is a genuine floor, not merely the best we managed, which is what makes it possible to say what a method has and has not learned rather than only which method won.

2. What we found

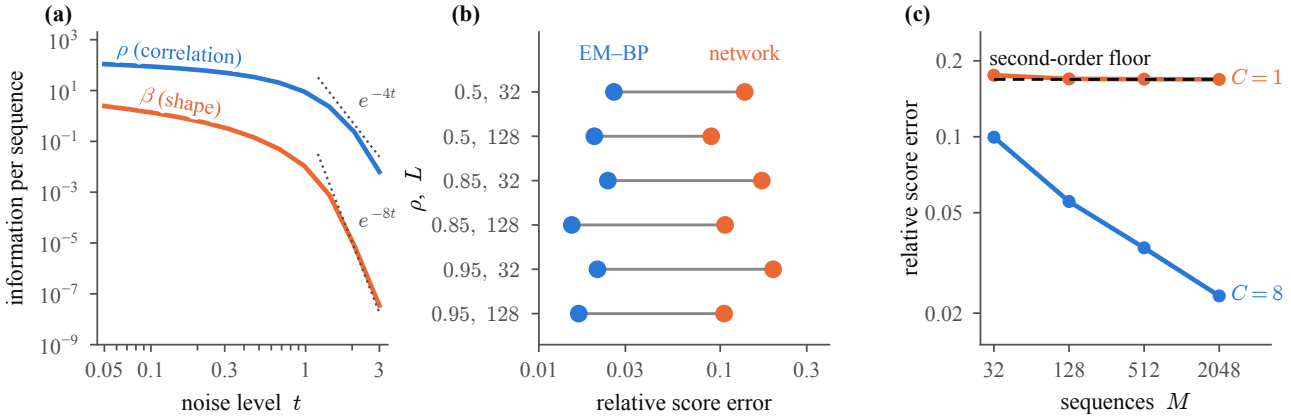


Figure 1: **(a)** What one noisy sequence still tells you about the two properties of the chain as noise rises: the correlation ρ and the shape β. Both fall exponentially but at different exponents; the dotted guides are the e^{-4t} and e^{-8t} the theory predicts. **(b)** Error of our estimator against the best network at six combinations of ρ and length L, on identical information. Lower is better. **(c)** Why structure helps: the dashed line is the best any method carrying only second-order information can do; a chain model restricted to Gaussian randomness ($C = 1$) sits on it at every sample size, while one free to represent non-Gaussian structure ($C = 8$) goes far below.

Noise destroys fine structure far faster than correlation, at an exact rate. This is the theoretical result, and panel (a) is it measured. Structure of each order decays at its own exponential rate under the noise, so the higher-order detail that makes data non-Gaussian vanishes first while plain correlation survives much longer. The two exponents differ by a factor of two, and the level at which they cross is a formula rather than a fitted curve. The practical consequence: above a computable noise level, everything a model can still learn about the data is second-order — and we can say where that level is.

Knowing the structure is worth 5–10× at equal information. Our estimator is ahead in all six regimes and in every one of the sixteen repetitions of each. The comparison is deliberately unflattering to us: the networks are re-screened and re-tuned separately inside each regime, while our method is denied the true score when choosing where to stop and selects on held-out evidence instead. The advantage is *smallest* where the chain carries the most non-Gaussian structure — the direction panel (a) predicts, so the experiment and the theory agree on a detail neither was fitted to.

ρ	L	LMMSE	EM–BP (ev.)	EM–BP (bp)	best net (C)	best net (B)	factor C / EM–BP (ev.)
0.5	32	0.265	0.026	0.024	0.137	0.129	5.7 [4.9, 6.7]
0.5	128	0.268	0.020	0.018	0.089	0.079	4.6 [4.1, 5.0]
0.85	32	0.167	0.024	0.021	0.170	0.156	8.3 [6.6, 10.3]
0.85	128	0.168	0.015	0.015	0.106	0.095	8.0 [6.7, 9.2]
0.95	32	0.086	0.021	0.019	0.196	0.128	10.3 [8.8, 11.9][†6]
0.95	128	0.087	0.017	0.016	0.105	0.094	6.4 [6.0, 6.9]

Table 2: Across correlation and length, 16 seeds, $M = 2048$. Columns: the second-order oracle; our estimator stopped by held-out evidence (ev.) and, for reference, by the true kernel (bp); the best network given the same single noisy view (C); the best network additionally given the *clean* sequence (B), an easier problem included to show what the restriction costs; and the factor between the C network and the evidence-selected estimator.

The advantage is fourth-order structure, and nothing else. The capacity experiment is the control. Restricted to a single Gaussian innovation, the chain model lands exactly on the second-order floor (0.168) and stays there however much data it is given; allowed richer innovations it goes far below. The only difference between those two arms is the freedom to represent non-Gaussian structure, so that is what the margin is made of. The same holds across fifteen further combinations of (ρ, L), with the restricted arm matching the theoretical floor to 0.0099 in the worst of them — which incidentally tests the grid, the message passing, the EM and the scoring simultaneously. It also holds with the non-Gaussianity reversed, on a chain *less* heavy-tailed than Gaussian: the factor there is 8.8 [7.4, 10.2].

Table 3: Schedule-pooled relative score error on all sites, mean over 16 seeds (s.e. in the text macros; the isotropic reference is 0.669 at every size). Every arm is scored on identical test bundles. The last column is the same capacity fitted to the innovation law itself—the C-component mixture closest in Kullback–Leibler divergence, an infinite-data reference for the family rather than a bound; † marks fits whose narrowest component is under two grid cells.

	sequences M				
kernel	32	128	512	2048	$M = \infty$
LMMSE floor	0.167	0.167	0.167	0.167	0.167
$C = 1$	0.174	0.168	0.168	0.167	0.167
$C = 2$	0.084	0.050	0.035	0.028	0.039
$C = 4$	0.093	0.053	0.037	0.025	0.0057
$C = 8$	0.099	0.055	0.036	0.023	0.0022†
$C = 16$	0.100	0.056	0.036	0.022	0.0007†

3. Where it stops working, and what the evidence rests on

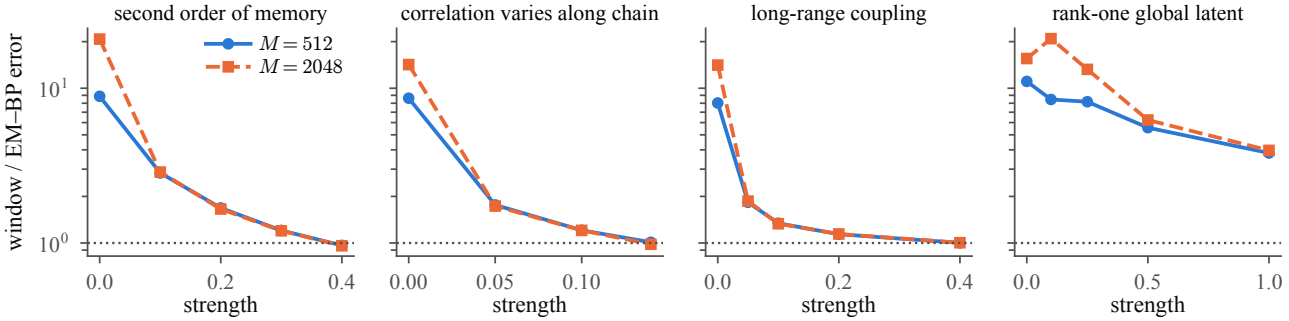


Figure 2: How wrong the first-order chain assumption may be before the advantage goes. Each panel pushes the data away from the assumed model in a different way and plots a network's error divided by ours; above the dotted line at 1 the misspecified chain estimator still wins. Every perturbation is Gaussian, so the true score of the *perturbed* model remains exactly computable and both sides are scored against it — we are not grading ourselves against our own assumption.

The boundary. Two departures reverse the ordering inside the range swept: adding a second order of memory, at a coefficient between 0.3 and 0.4, and letting the correlation vary along the sequence, between 0.1 and 0.14. Two others never reverse it — a rank-one global latent is absorbed into the fitted correlation, and a long-range coupling decays to a tie. The two sample sizes also separate the two ways of being wrong: at zero perturbation the error halves when the data quadruples, which is estimation error; past break-even it does not move at all, which is bias, and no amount of data will fix it.

What the numbers rest on. Every experiment behind this page is frozen: its outputs are hash-pinned and its configuration was registered before it ran. Each number here is generated from those outputs by a tool and then recomputed independently by a second tool that fails if the two disagree, and the figures check themselves against the manuscript's own numbers and refuse to be written if a value does not match. One experiment, a sample-complexity measurement, is still running on the cluster; nothing on this page depends on it.

One correction on the record. The capacity sweep in the earlier archival study is void. Its training set was drawn in a way that re-created the random generator for each sequence, so every configuration was in fact fitted to a single repeated sequence. The earlier explanation offered for that study's conclusion — a stopping-rule asymmetry — is withdrawn with it. The capacity evidence above was computed independently and is unaffected.